Muirfield High School

YEAR 9 - MATHEMATICS

Topic Test:
Measurement, Perimeter, Area and Surface Area

Name: ---

Class: --------------------   Mark: -------------

Note: ***All necessary working must be shown. Marks will be awarded for necessary working out.***

Section I (5.2 and 5.3)

Fill in the gaps: **10 Marks**

1. To convert from kilometres to metres, you __________________ by 1000.
2. The ________________ of a shape is the distance around the shape.
3. The formula C = __________________ is used to find the circumference (perimeter) of a circle.
4. The _____________________ of a shape is a measure of the amount of surface enclosed by the shape.
5. Area of a circle can by found using the formula A = _______________________.
6. A _____________________ is a portion of a circle formed by two radii and the arc of the circle between them.
7. To find the area of a sector, multiply the formula for the area of a circle by the _________________ of the circle formed by the sector.
8. Prisms are 3-dimentional figures, which have uniform _________________.
9. To find the surface area of a prism, find the area of each ____________ and add them together.
10. The surface of a cylinder is made up of two _________________ and a ____________________.

perimeter	circles	face
multiply	rectangle	fraction
πr^2	$2\pi r$	sector
area	cross-sections	

1. Complete the following conversions: **4 Marks**

(a) 3.7 m = __________ mm

(b) 8.523 g = __________ kg

(c) 4 hr = __________ min

(d) 3.58 t = __________kg

2. Use the '*Degrees, Minutes, Seconds*' button on your calculator to write the following in hours and minutes: **2 Marks**

(a) 2.5 hr = __________

(b) 1.2 hr = __________

3. Write the following in hours: **2 Marks**

(a) 3 hr 54 min = __________

(b) 2 hr 24 min = __________

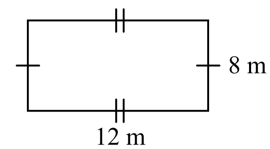
4. With the aid of a calculator perform the following operations: **2 Marks**

(a) 2 hr 10 min + 4 hr 50 min = _______

(b) 5 hr 28 min - 2 hr 45 min = _______

5. Find the perimeter of the following shapes: **3+6 Marks**

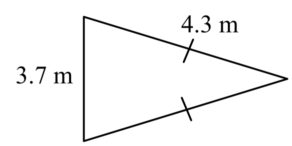
(a)



____

____

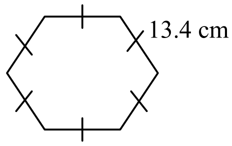
(b)



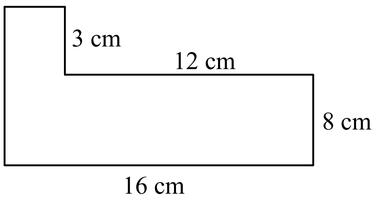
____

____

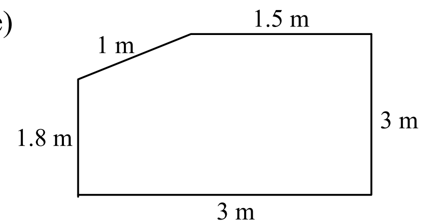
(c)



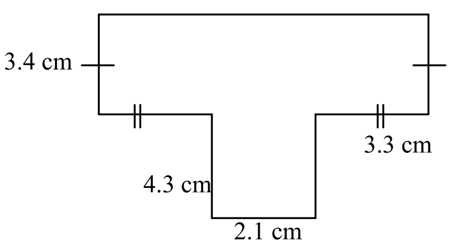
(d)



(e)

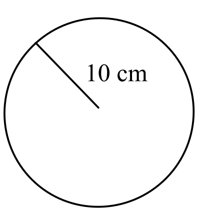


(f)



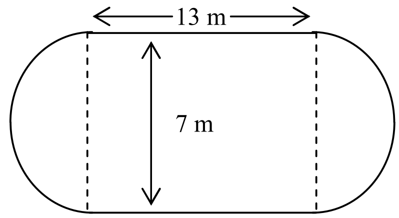
6. Calculate the circumference of the following circle (correct to 2 decimal places).

3 Marks



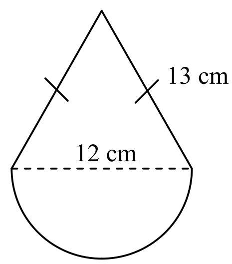
7. Find the perimeter of these shapes (correct to 1 decimal place: **4 Marks**

(a)

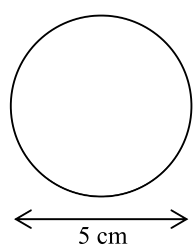


4 Marks

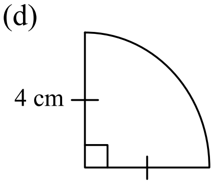
(b)



(c) **3 Marks**



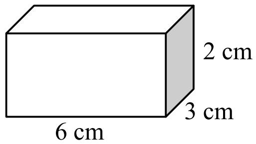
(d) **3 Marks**



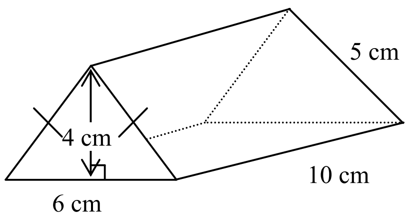
8. Calculate the total ***surface area*** of the following shapes:

4 Marks

a.

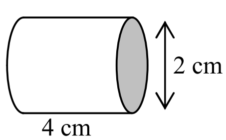


b.



4 Marks

c. (Hint: Use the equation $S.A. = 2\pi r^2 + 2\pi rh$)



4 Marks

9. A circular running track has a diameter of 80 metres.

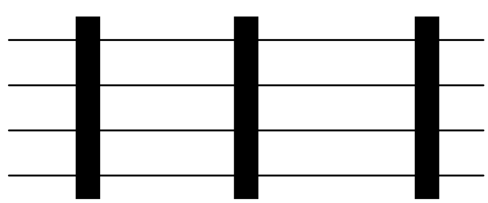
(a) How far will Peter run with each completed lap? **2 Marks**

(b) How many laps will Peter need to run to complete 1000 m? **1 Marks**

10. Find the perimeter of a right-angled triangle with hypotenuse length 17 cm and one other side of 8 cm by first drawing a diagram.

3 Marks

11. A farmer decided to fence a **400 m** by **50 m** paddock with a four-strand wire fence.



(a) Find the perimeter of the paddock

2 Marks

(b) Find the total cost of the wire required given that single strand wire costs 12.4 cents per metre. **2 Marks**

End.

Muirfield High School

YEAR 9 - MATHEMATICS

Topic Test: Volume

Name: ---

Class: ---------------------   Mark: --------------

Note: ***All necessary working must be shown. Marks will be awarded for necessary working out.***

Section II (5.3 only)

1. Convert the following volumes to the units specified. **3 Marks**

(a) 5.23 litres = _________ cm^3

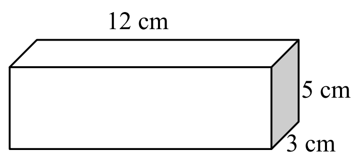
(b) 6000 cm^3 = _________ mL

(c) 5 m^3 = __________litres

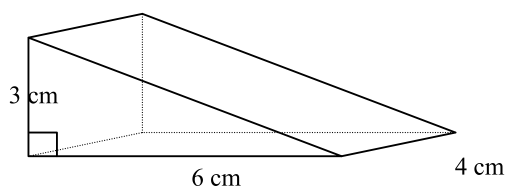
2. Find the volume of the following:

2+3+3 Marks

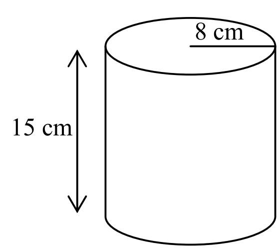
a.



b.



c.

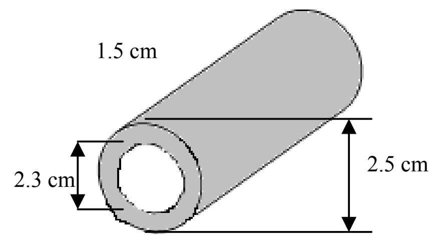


3.

a. Calculate the volume of metal in the pipe shown, correct to 1 decimal place.

4 Marks

b. Calculate the weight of the pipe if 1 cm^3 of metal weighs 5.8 g. Answer correct to 1 decimal place. **2 Marks**



Muirfield High School

YEAR 9 - MATHEMATICS

Topic Test:
Measurement, Perimeter, Area and Surface Area

Name: -----------**SOLUTIONS**----------

Class: --------------------- **Mark:** ------- 65

Note: ***All necessary working must be shown. Marks will be awarded for necessary working out.***

Section I (5.2 and 5.3)

Fill in the gaps: **10 Marks**

perimeter	circles	face
multiply	rectangle	sector
πr^2	$2\pi r$	
area	cross-sections	

1. To convert from kilometres to metres, you **multiply** by 1000. ✓
2. The **perimeter** of a shape is the distance around the shape. ✓
3. The formula C = _**$2\pi r$**_ is used to find the circumference (perimeter) of a circle. ✓
4. The _**area**_ of a shape is a measure of the amount of surface enclosed by the shape. ✓
5. Area of a circle can by found using the formula A = _**πr^2**_. ✓
6. A ____**sector**____ is a portion of a circle formed by two radii and the arc of the circle between them. ✓
7. Prisms are 3-dimentional figures, which have uniform ___**cross-sections**___. ✓
8. To find the surface area of a prism, find the area of each ___**face**___ and add them together. ✓
9. The surface of a cylinder is made up of two _ circles ____**circles**____ and a _**rectangle**_. ✓✓

1. Complete the following conversions: **4 Marks**

(a) 4 hr = ______**240**______ min ✓

(b) 3.58 t = ____**3580**____kg ✓

(c) 3.7 m = __**3700**__ mm ✓

(d) 8.523 g = ___**0.008523**_____ kg ✓

2. Use the '*Degrees, Minutes, Seconds*' button on your calculator to write the following in hours and minutes: **2 Marks**

(a) 2.5 hr = ___**2hr 30 min**___ ✓

(b) 1.2 hr = ___**1hr 12 min**___ ✓

3. Write the following in hours: **2 Marks**

(a) 3 hr 54 min = __**3.9 hr**__ ✓

(b) 2 hr 24 min = __**2.4 hr**___ ✓

4. With the aid of a calculator perform the following operations: **2 Marks**

(a) 2 hr 10 min + 4 hr 50 min =

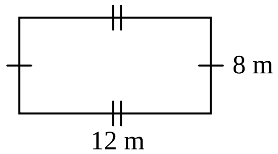
__**7 hr**__ ✓

(b) 5 hr 28 min - 2 hr 45 min =

___**2 hr 43 min**___ ✓

5. Find the **perimeter** of the following shapes: **1+1+1+1+2+2 Marks**

(a)

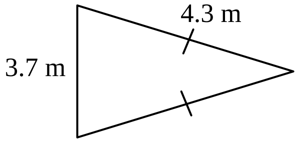


_**P = 12 x 12 + 8 x 2 = 40 m**______ ✓

____

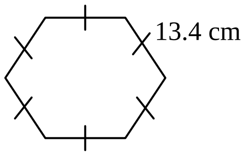
21

(b)



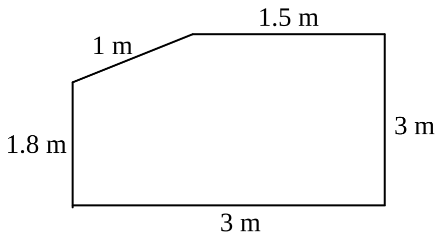
P = 4.3 x 2 + 3.7 = 12.3 m ✓

(c)



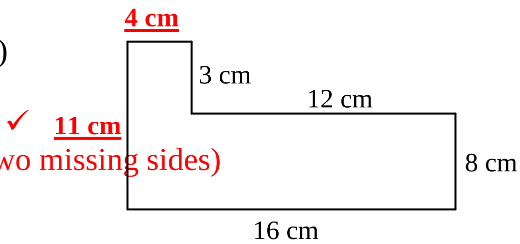
P = 13.4 x 6 = 80.4 cm ✓

(d)



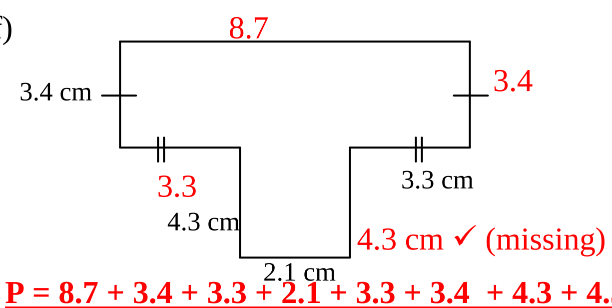
P = 1.8 + 1 +1.5 + 3 + 3 = 10.3 m ✓

(e)



P = 11 + 4 + 3 + 12 + 8 + 16 = 54 cm ✓

(f)

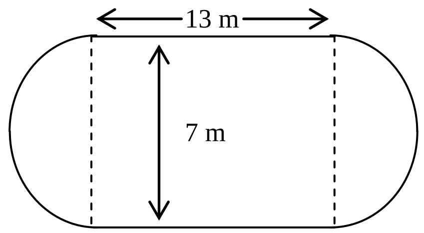


P = 8.7 + 3.4 + 3.3 + 2.1 + 3.3 + 3.4 + 4.3 + 4.3

= 32.8 cm ✓

6. Find the **perimeter** of these shapes (correct to 1 decimal place: **4 Marks**

(a)



2 semi circles = 1 circle

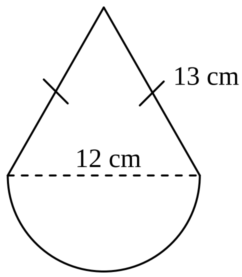
r = 3.5 m ✓

P =2 x π x 3.5 + 13 x 2 = 47.9914858 m ✓✓

≑ 48.0 m ✓

4 Marks

(b)

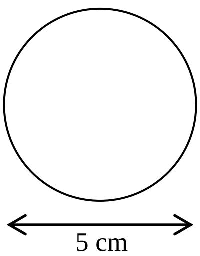


P = semi circle + 13 x 2 ✓

= π x 6 + 13 x 2 = 44.8495 … cm ✓✓

≑ 44.8 cm ✓

(c) **3 Marks**



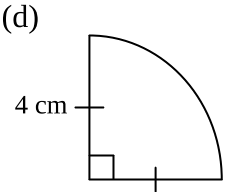
C = 2πr

= 2 x π x 2.5 ✓

= 15.70796…… ✓

≑15.7 cm ✓

(d) **3 Marks**



P = 2πr ÷ 4 + 4 x 2

= 2 x π x 4 ÷ 4 + 8

✓

=14.28318…… ✓

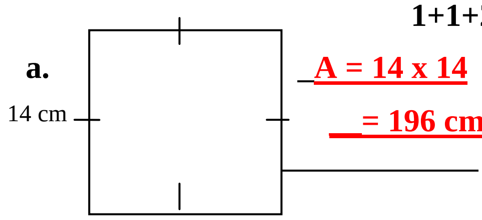
≑14.3 cm ✓

21

7. Find **area** of the following shapes.

1+1+2 Marks

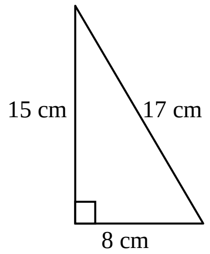
a.



A = 14 x 14

= 196 cm^2 ✓

b.

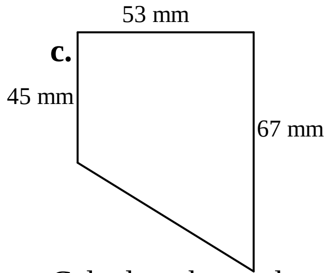


A= ½ x b x h

= ½ x 8 x 15

=60 cm^2 ✓

c.



A= ½(a + b) x h

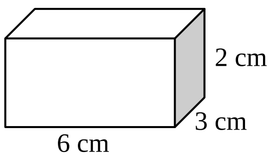
= ½ (45 + 67) x 53✓

= 2968 mm^2 ✓

8. Calculate the total ***surface area*** of the following shapes:

4 Marks

a.

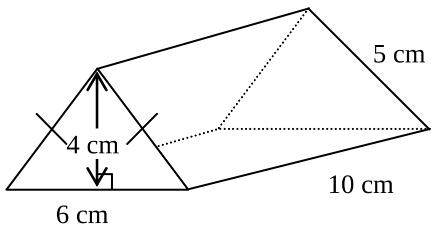


Surface Area =

= 2 x [(6 x 2) + (3 x 2) + (6 x 3)] ✓✓✓

= 72 cm^2 ✓

b.



4 Marks

Front and Back ✓

= (½ x 6 x 4) x 2 = 24 cm^2

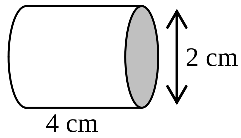
Left and Right ✓

= 5 x 10 x 2 = 100 cm^2

Bottom = 10 x 6 = 60 cm^2 ✓

T.S.A. = 24 + 100 + 60 = 184 cm^2 ✓

c. (Hint: Use the equation $S.A. = 2\pi r^2 + 2\pi rh$)



3 Marks

S. A. = 2 x π x 1^2 + 2 x π x 1 x 4 ✓✓

= 31.41592654 cm^2 ✓

(rounded decimal numbers are acceptable)

9. A circular running track has a diameter of 80 metres.

(a) How far will Peter run with each completed lap? **1 Mark**

1 lap = π x 80 or 2 x π x 40

= 251.3274123

≑215.33 m ✓

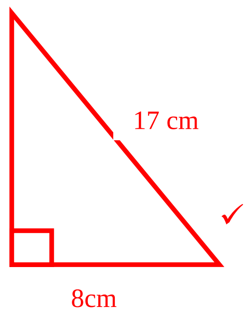
(b) How many laps will Peter need to run to complete 1000 m? **1 Mark**

1000 ÷ 251.33

≑4 laps ✓

10. Find the perimeter of a right-angled triangle with hypotenuse length 17 cm and one other side of 8 cm by first drawing a diagram.

3 Marks

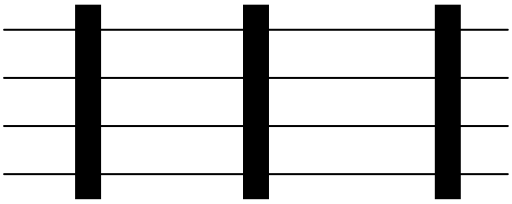


$x = \sqrt{17^2 - 8^2}$ = 15 cm ✓

P = 17 + 15 + 8 = 40 cm ✓

20

11. A farmer decided to fence a **400 m** by **50 m** paddock with a four-strand wire fence.



(a) Find the perimeter of the paddock

1 Mark

400 + 400 + 50 + 50 = 900 m ✓

(b) Find the total cost of the wire required given that single strand wire costs 12.4 cents per metre. **2 Marks**

Length of wire = 900 x 4 = 3600 m ✓

Cost of wire = 3600 x 12.4

= 44640 cents

= $446.40 ✓

End.

3

Muirfield High School

YEAR 9 - MATHEMATICS

Topic Test: Volume

Name: SOLUTIONS

Class: --------------------- **Mark:** --------------

Note: ***All necessary working must be shown. Marks will be awarded for necessary working out.***

Section II (5.3 only)

1. Convert the following volumes to the units specified. **3 Marks**

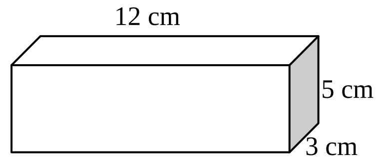
(a) 5.23 litres = _5230_ cm^3 ✓

(b) 6000 cm^3 = 6000 mL ✓

(c) 5 m^3 = 5000 litres ✓

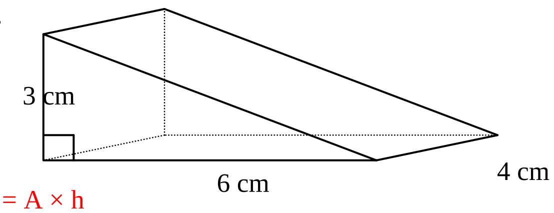
2. Find the volume of the following: **2+3+3 Marks**

a.



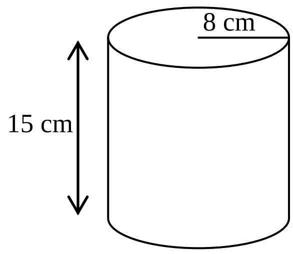
$V = 12 \times 5 \times 3 = 180\ cm^2$
✓ ✓

b.



$V = A \times h$
$= (½ \times 3 \times 6) \times 4 = 36\ cm^3$
✓ ✓ ✓

c.

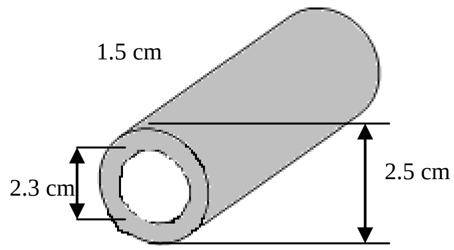


$V = A \times h = \pi r^2 h$
$= \pi \times 8^2 \times 15$ ✓ ✓
$= 3015.93\ cm^3$ ✓

3.

a. Calculate the volume of metal in the pipe shown, correct to 1 decimal place. **4 Marks**

b. Calculate the weight of the pipe if 1 cm^3 of metal weighs 5.8 g. Answer correct to 1 decimal place. **2 Marks**



a. Volume of metal
= Volume of outer cylinder – Volume of inner cylinder ✓
$= \pi R^2 h - \pi r^2 h$
$= \pi \times 1.25^2 \times 1.5 - \pi \times 1.15^2 \times 1.5$
✓ ✓
$= 1.1\ cm^3$ (approx.) ✓

b. Weight of the pipe = 1.1×5.8 g ✓
= 6.38 g ✓
(If they use calculator ans key from part (a) = 6.6 g ✓)

/17